PARISHRAM 2.0

CLASS 12TH 2026

MATRIX

MATHS LECTURE-4

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TOPICS TO BE COVERED

Matrix Multiplication

1. Pollwa Questions
2. Matrix Multiplication Practice Questions



POLL QUESTION

If $\begin{bmatrix} x & 2 & 0 \end{bmatrix} \begin{bmatrix} 5 \\ -1 \\ x \end{bmatrix} = \begin{bmatrix} 3 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ x \end{bmatrix}$, then value of x is:



A -1

B 0

C 1

D 2

$$\begin{bmatrix} x & 2 & 0 \end{bmatrix} \begin{bmatrix} 5 \\ -1 \\ x \end{bmatrix} = \begin{bmatrix} 3 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ x \end{bmatrix}$$

$$5x + 2(-1) + 0(x) = 3(-2) + 1(x)$$

$$5x - 2 = -6 + x$$

$$5x - 1x = -6 + 2$$

$$4x = -4$$

$$x = -\frac{4}{4}$$

$$\boxed{x = -1}$$

POLL QUESTION

Given that $\begin{bmatrix}1 & x\end{bmatrix}\begin{bmatrix}4 & 0\\ -2 & 0\end{bmatrix} = 0$, the value of x is:

A −4

B −2

C 2

D 4

$$\begin{bmatrix}1 & x\end{bmatrix}\begin{bmatrix}4 & 0\\ -2 & 0\end{bmatrix} = 0$$

$$\begin{bmatrix}1(4)+x(-2) & 1(0)+x(0)\end{bmatrix} = 0$$

$$\begin{bmatrix}4-2x & 0\end{bmatrix} = \begin{bmatrix}0\end{bmatrix}$$

$$4-2x=0$$

$$4=2x \rightarrow x=4/2$$

$$\boxed{x=2}$$



POLL QUESTION

If $A = \begin{bmatrix} 2 & -1 & 3 \\ -4 & 5 & 1 \end{bmatrix}_{2\times3}$ and $B = \begin{bmatrix} 2 & 3 \\ 4 & -2 \\ 1 & 5 \end{bmatrix}_{3\times2}$

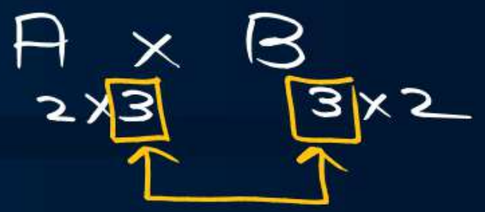


A Only AB is defined

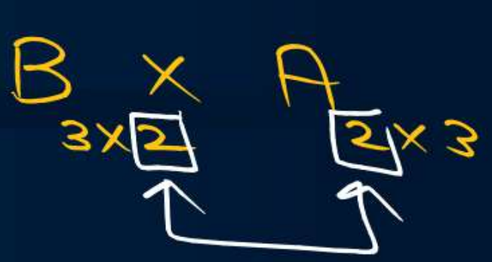
B Only BA is defined

C AB and BA both are defined

D AB and BA both are not defined



AB ✓



BA ✓

$$A = \begin{bmatrix} 3 & 4 \\ 5 & 7 \end{bmatrix}$$

$A^2 = \begin{bmatrix} 9 & 16 \\ & \end{bmatrix}$ ✗

Wrong

$A^2 = A \times A$

$$A^2 = \begin{bmatrix} 3 & 4 \\ 5 & 7 \end{bmatrix}\begin{bmatrix} 3 & 4 \\ 5 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} 3(3)+4(5) & 3(4)+4(7) \\ 5(3)+7(5) & 5(4)+7(7) \end{bmatrix} = \begin{bmatrix} 29 & 40 \\ 50 & 69 \end{bmatrix}$$

$A^3 = A^2 \times A$

$$= \begin{bmatrix} 29 & 40 \\ 50 & 69 \end{bmatrix}\begin{bmatrix} 3 & 4 \\ 5 & 7 \end{bmatrix}$$

Comment

POLL QUESTION

For the matrix $X = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$, $(X^2 - X)$ is



A. 2I

B. 3I

C. I

D. 5I

$X^2 = X \cdot X$

$= \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

$= \begin{bmatrix} 0+1+1 & 0+0+1 & 0+1+0 \\ 0+0+1 & 1+0+1 & 1+0+0 \\ 0+1+0 & 1+0+0 & 1+1+0 \end{bmatrix}$

$X^2 = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$

$$X^2 - X$$

$$= \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} - \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$= 2\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 2I \rightarrow \boxed{a}$$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

POLL QUESTION

If $A = [2 \quad -3 \quad 4], B = \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix}, X = [1 \quad 2 \quad 3]$ and $Y = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$ then AB + XY equals

8 + 20 = 28

A [28]

B [24]

C 28

D 24

$AB = \begin{bmatrix} 2 & -3 & 4 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix}$

$AB = [2(3) + (-3)(2) + 4(2)]$

$= [6 - 6 + 8]$

$AB = 8$

$XY = [1 \quad 2 \quad 3] \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$

$= [2 + 6 + 12]$

$= [8 + 12]$

$= [20]$



QUESTION

$(A+B)(A-B) \neq A^2 - B^2$

If $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, then show that $(A + B)(A - B) \neq A^2 - B^2$.



$$A+B = \begin{bmatrix} 0 & 0 \\ 2 & 1 \end{bmatrix}$$

$$A-B = \begin{bmatrix} 0 & 2 \\ 0 & 1 \end{bmatrix}$$

$$(A+B)(A-B) = \begin{bmatrix} 0 & 0 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 0 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 5 \end{bmatrix}$$

$$A^2 = A \times A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0+1 & 0+1 \\ 0+1 & 1+1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

$$B^2 = B \times B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$A^2 - B^2 = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$

QUESTION

For the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$, show that $A^3 - 6A^2 + 5A + 11I = 0$

✓ ✓ ✓ ✓



$$A^2 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1+1+2 & 1+2-1 & 1-3+3 \\ 1+2+(-6) & 1+4+3 & 1-6-9 \\ 2-1+6 & 2-2-3 & 2+3+9 \end{bmatrix} = \begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix}$$

$A^3 = A^2 \times A$

$$= \begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 4+2+2 & 4+4-1 & 4-6+3 \\ -3+8-28 & -3+16+14 & -3-24-42 \\ 7-3+28 & 7-6-14 & 7+9+42 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 7 & 1 \\ -23 & 27 & -69 \\ 32 & -13 & 58 \end{bmatrix}$$

-27
-42

Comment

THANK YOU